

## Objective

The objective of the Day 5 project is to develop an interactive **Number Guessing Game** using Python. The project focuses on understanding random number generation, loops, conditional statements, user input, and repeated interaction. The program generates a random number within a specified range and allows the user to continue entering guesses until the correct number is identified. The project also helps improve problem-solving skills and provides practical experience in developing an interactive Python application.

## Key Features

The Number Guessing Game provides an interactive interface where the computer generates a random number between 1 and 100 and the user attempts to identify it by entering different guesses. After every guess, the application provides a clear hint indicating whether the entered number is too high or too low. An attempt counter keeps track of the number of guesses made by the user, and the game automatically stops when the correct number is found. A congratulatory message displays the successful result along with the number of attempts taken. The application also provides a **Start New Game** option that generates a new random number and resets the attempt counter.

## Technologies Used

The project is developed using **Python** as the primary programming language. Python's built-in **random module** is used to generate the secret number using `random.randint()`. **Streamlit** is used to create a clear and interactive graphical interface for the game. Conditional statements such as `if`, `elif`, and `else` are used to compare the user's guess with the generated number and provide appropriate hints. Python variables and session state are used to maintain the generated number, attempt count, and game status during user interaction.

## Learning Outcome

This project provides practical understanding of **random number generation, conditional logic, user interaction, state management, and repeated program execution**. It also demonstrates how basic Python programming concepts can be combined to create an interactive application. Through this project, the concepts of the random module, comparison operators, conditional statements, input handling, and attempt counting are applied in a real-world programming scenario.

C:\Users\shamb\OneDrive\Desktop\VEDA internship\DAY5>

```
app.py > ...
1 import streamlit as st
2 import random
3
4 st.set_page_config(
5     page_title="Number Guessing Game",
6     page_icon="🎲",
7     layout="centered"
8 )
9 st.markdown("""
10 <style>
11     .main {
12         padding-top: 2rem;
13     }
14
15     .game-title {
16         text-align: center;
17         font-size: 42px;
18         font-weight: 700;
19         margin-bottom: 5px;
20     }
21
22     .subtitle {
23         text-align: center;
24         font-size: 18px;
25         margin-bottom: 30px;
26     }
27
28     .info-card {
29         padding: 20px;
30         border-radius: 12px;
31         text-align: center;
32         margin-bottom: 15px;
33         border: 1px solid #ddd;
34     }
35
36     .attempt-number {
37         font-size: 28px;
38         font-weight: bold;
39     }
40
41     .success-box {
42         padding: 20px;
43         border-radius: 12px;
44         text-align: center;
45         background-color: #d1fae5;
46         border: 1px solid #10b981;
47     }
48
49     .hint-box {
50         padding: 18px;
51         border-radius: 12px;
52         text-align: center;
53         font-size: 18px;
54         margin-top: 15px;
55     }
56
57     .footer {
58         text-align: center;
59         margin-top: 40px;
60         font-size: 14px;
61     }
62 </style>
63 """, unsafe_allow_html=True)
64
65 if "target_number" not in st.session_state:
66     st.session_state.target_number = random.randint(1, 100)
67
68 if "attempts" not in st.session_state:
69     st.session_state.attempts = 0
70
71 if "game_over" not in st.session_state:
```

```
PS C:\Users\shamb\OneDrive\Desktop\VEDA internship\DAY5> python -m streamlit run app.py
2026-09-24 17:59:39.833 Uvicorn server started on :::8501
```

You can now view your Streamlit app in your browser.

Local URL: <http://localhost:8501>

Network URL: <http://192.168.1.4:8501>

Help agents write better Streamlit apps?

Install the official Streamlit skills by running `streamlit skills` in your terminal.

<http://localhost:8501>



# Number Guessing Game

Guess the secret number between **1** and **100**!

 Number Range

**1 - 100**

 Attempts

**10**

Enter your guess:

84

- +

 Check My Guess



## Congratulations!

You guessed the correct number:

**84** 

You found it in **11** attempts.

▼  How to Play

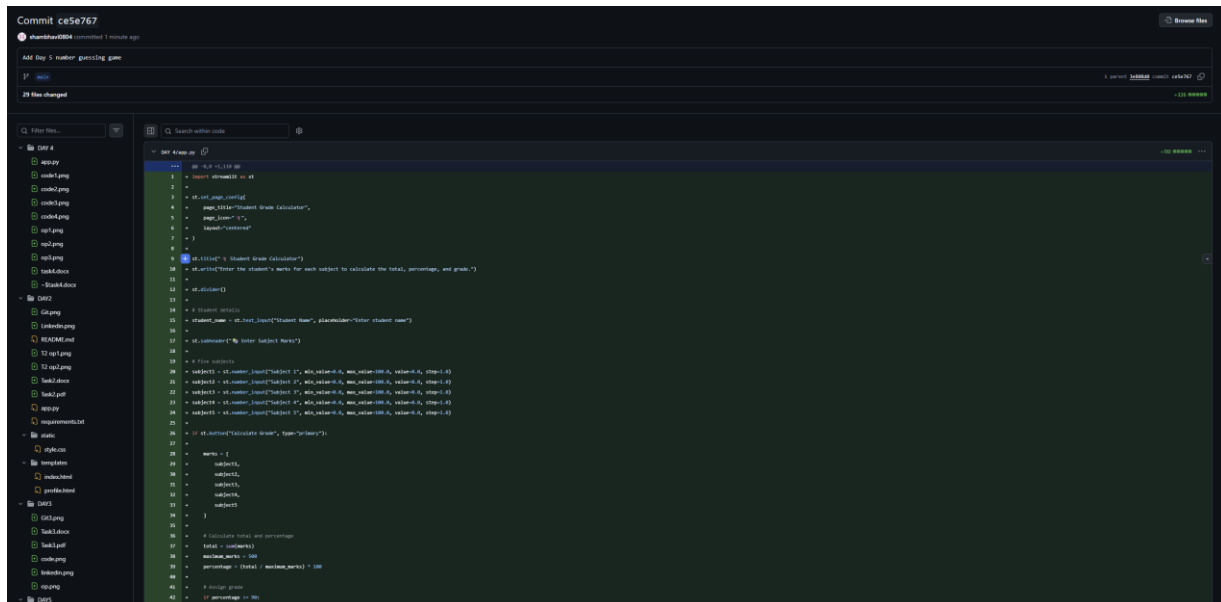
1. The computer generates a random number between **1** and **100**.
2. Enter your guessed number.
3. Click **Check My Guess**.
4. The game tells you whether your guess is **too high** or **too low**.
5. Continue guessing until you find the correct number.
6. Your total number of attempts is displayed.



Python Programming Track | Day 5

Built using Python, Random Module & Streamlit

Github: <https://github.com/shambhavi0804/VEDA-internship/commit/ce5e767c52bc94eb6828e70695c7cb60187ab25d>



The screenshot shows a GitHub commit page for the file 'number\_guessing\_game.py'. The commit message is 'Add Day 5 number\_guessing\_game'. The file is 17 lines long. The code is a Python script that implements a number guessing game. It uses the 'random' module to generate a random number between 1 and 100. The user is prompted to enter a guess, and the program checks if the guess is correct. If the guess is incorrect, it tells the user if the number is too high or too low. The game continues until the user guesses the correct number. The code is as follows:

```
1 import random
2
3 # Set up game config
4 # Generate a random number between 1 and 100
5 # Set up game variables
6 # Set up game logic
7 # Set up game display
8 # Set up game control
9
10 # Set up game config
11 # Generate a random number between 1 and 100
12 # Set up game variables
13 # Set up game logic
14 # Set up game display
15 # Set up game control
16
17 # Set up game config
18 # Generate a random number between 1 and 100
19 # Set up game variables
20 # Set up game logic
21 # Set up game display
22 # Set up game control
```

LinkedIn: <https://lnkd.in/p/drAFgB4N>

